REPORT 015: EPOCH 015 — KONDUKTIVA MUSICALENVIRONMENT BRIDGE & LIVE 8-LANE IOI RHYTHM VISUALIZER

2026-11-04 01:37:35 | Implementation of js/konduktiva-bridge.js and src/konduktiva_view.rs streaming QuantizedMap IOI rhythm rows, mask maps, ladder rungs, and note triggers over UDP OSC into a live 8-lane egui Rhythm Activity & Step Visualizer

Operator: Antigravity (Google DeepMind) · Follows: report-014

Prompt

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Epoch 15 of 20: Deepen the Konduktiva integration across `js/konduktiva-bridge.js`
↪  and `src/konduktiva_view.rs`: connect Konduktiva `MusicalEnvironment` and
↪  rhythm-map ladders (`/home/renick/claude/konduktiva` and
↪  `/home/renick/claude/8operc-konduktiva`) to stream real-time per-player IOI
↪  (inter-onset interval) rhythm rows, mask maps, density graph positions, and note
↪  trigger events over OSC into the Rust `egui` cockpit. Render a live 8-lane Rhythm
↪  Activity & Step Visualizer in `egui` so the performer can see upcoming rhythm
↪  density changes and voice triggers alongside physical controller movements. Add
↪  automated tests (with `CARGO_BUILD_JOBS=2`), git commit all changes, and compile
↪  `reports/report-015.html` from `reports/report-015.md` via
↪  `/home/renick/claude/html-report-generator/build-report.sh --pdf` and commit.
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TL;DR

Implemented the Konduktiva MusicalEnvironment & Rhythm-Map Ladder Bridge in js/konduktiva-bridge.js and the live 8-lane Rhythm Activity & Step Visualizer in src/konduktiva_view.rs: (1) models and extracts Konduktiva sparse-to-dense QuantizedMap rhythm ladders (5 rungs per voice lane), IOI (inter-onset interval) rows, maskMap flags, velocityMap values, and deterministic per-hit variation depth contours; (2) streams real-time per-lane rhythm rows (/nkeditor/konduktiva/lane//rhythm) and note triggers (/nkeditor/konduktiva/lane//trigger) over UDP OSC; and (3) renders an interactive 8-row custom vector IOI step visualizer in egui with moving playhead and trigger flashes. Verified with 57 passing Rust tests and 12 passing Node.js tests (69 total).

Plain-Language Overview: What Changed and Why It Matters

In traditional step sequencers, a drum pattern is just a fixed 16-step grid of on/off boxes. **Konduktiva** (`/home/renick/claude/konduktiva` and `/home/renick/claude/8operc-konduktiva/src/pulsed-core-r` works differently: instead of a rigid grid, each percussion voice (`Player`) reads a **QuantizedMap rhythm ladder** (`rhythmMap`)—a stack of 5 pre-calculated rhythmic rows ordered from sparse (few hits per bar) to dense (rapid 16th- and 32nd-note subdivisions). Each row stores the exact beat positions where notes start (`onset_beats`) and the **Inter-Onset Interval (IOI)**—the time gap in beats until the next hit (`ioi_values`). When a performer turns a physical density knob on the Korg `nanoKONTROL2` (`CC 16..23`), Konduktiva climbs up or down this 5-rung ladder (`selected_rung_index` out of `total_rungs`).

Before Epoch 15, the performer could see the knob's numeric density multiplier (`1.00x` to `5.00x`), but could not *see* the actual rhythmic pattern on screen or watch upcoming hits approach the playhead.

In Epoch 15, we connected Konduktiva's internal rhythm engine directly to the Rust `egui` cockpit through `js/konduktiva-bridge.js` and `src/konduktiva_view.rs`:

1. **Per-Player IOI Rhythm Row & Ladder Extraction (`js/konduktiva-bridge.js`)**: Extracts each voice lane's active `QuantizedMap` rung (`selectedRungIndex`, `totalRungs`), its `onsetBeats` and `ioiValues`, its `maskFlags` (`maskMap` indicating which steps are silenced or muted), its `velocities` (`velocityMap`), and its deterministic per-hit timbre variation depth (`VARIATION_DEPTH_RANGES`).
2. **Real-Time UDP OSC Streaming (`KonduktivaBridge`)**: Whenever a physical knob changes a lane's density or a Konduktiva `Player` fires a note, `KonduktivaBridge` streams `/nkeditor/konduktiva/lane//rhythm` and `/nkeditor/konduktiva/lane//trigger` packets over UDP OSC to the Rust application.
3. **Live 8-Lane Rhythm Activity & Step Visualizer (`src/konduktiva_view.rs`)**: Renders an 8-row custom vector timeline in `egui` (`render_konduktiva_rhythm_visualizer`) showing proportional IOI blocks across the loop `keyspan` (`4`, `8`, or `16` beats), muted/masked steps, the active ladder rung badge (`Rung 3/5`), a moving white playhead line, and the latest triggered MIDI note, velocity, IOI, and FM timbre patch.

1. Plain-Language to Technical Component Mapping

The table below maps each musical and visual concept in Epoch 15 to its exact implementation in `js/konduktiva-bridge.js` and `src/konduktiva_view.rs`:

Plain-Language Concept	What It Means Musically	Node.js Component ('js/konduktiva-bridge.js')	Rust Component ('src/konduktiva_view.rs')
Inter-Onset Interval (IOI)	The exact beat duration from one drum hit to the next (`nextBeat - currentBeat`) inside a looping bar	`row.keys` (`onsetBeats`) & `row.values` (`ioiValues`)	`KonduktivaLaneRhythmRow::onset_beats` & `ioi_values`
5-Rung Rhythm Ladder (`rhythmMap`)	Stack of 5 rhythmic patterns per voice from Rung 1 (sparse) to Rung 5 (maximum subdivision)	`buildRhythmMapLadderData`, `selectLadderRung`	`build_rhythm_map_ladder`, `QuantizedRhythmRung`, `update_from_density`
Step Mask Map (`maskMap`)	Per-step boolean flag (`true` = silent/masked, `false` = audible hit) driven by pattern masks and lane Mute/Solo	`extractPlayerRhythmRow` (`maskFlags`)	`KonduktivaLaneRhythmRow::mask_flags`, `active_step_at_beat`
Per-Hit Variation Contour	Deterministic dual-sine wave (`VARIATION_DEPTH_RANGES`) that breathes timbre variation depth between accents	`VARIATION_DEPTH_RANGES`, `computeVariationDepth`	`VARIATION_DEPTH_RANGES`, `compute_variation_depth`
Real-Time OSC Telemetry Stream	UDP OSC messages carrying updated IOI rows and per-hit note triggers from Node.js to Rust	`KonduktivaBridge.syncLane`, `emitNoteTrigger`, `stepBeatWindow`	`KonduktivaTelemetryState::apply_lane_rhythm_json`, `record_note_trigger`

Plain-Language Concept	What It Means Musically	Node.js Component (‘js/konduktiva-bridge.js‘)	Rust Component (‘src/konduktiva_view.rs‘)
8-Lane Step Visualizer (egui)	8-row vector timeline painting beat gridlines, proportional IOI blocks, active step highlight, and playhead	Driven by KonduktivaBridge & Op8PerformanceModel	render_konduktiva_rhythm_visualizer

2. Per-Family Variation Depth Ranges (‘VARIATION_DEPTH_RANGES‘)

Both js/konduktiva-bridge.js and src/konduktiva_view.rs implement the exact per-family variation depth contours defined in /home/renick/claude/8operc-konduktiva/src/pulsed-core-rhythm.js:

Lane Index	‘nkEditor4‘ Family	‘Konduktiva‘ Family	Variation Depth Min	Variation Depth Max	Native 4-Beat Onsets (Rung 3)
0	kick	kicks	0.45	0.95	0.0, 1.5, 2.0, 3.25
1	snare	snares	0.82	1.45	1.0, 3.0
2	hat_closed	hats	0.72	1.55	0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5
3	hat_open	toms	0.82	1.50	0.75, 2.75
4	clap	bells	0.78	1.50	1.0, 2.5, 3.0
5	tom	metals	0.70	1.28	0.5, 1.75, 3.5
6	rim	noise	0.75	1.48	0.25, 1.25, 2.25, 3.25
7	cymbal	fx	0.72	1.50	0.0, 2.0

Tests

QA

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**Total Tests**: `69` passing (`57` Rust tests + `12` Node.js tests), `0` failing, `0` compiler warnings.

**Safety & Resource Discipline**: All builds and tests executed with `CARGO_BUILD_JOBS=2` in headless/dry-run mode without launching blocking GUI windows or starting live audio playback.

**Verdict**: **PASS**.
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Files touched

- ‘js/konduktiva-bridge.js‘ (Created — 405 lines): Implemented ‘VARIATION_DEPTH_RANGES‘, ‘buildRhythmMapLadderData‘, ‘selectLadderRung‘, ‘computeVariationDepth‘, ‘extractPlayerRhythmRow‘, and ‘KonduktivaBridge‘ with real-time UDP OSC streaming of IOI rhythm rows and note triggers.
- ‘src/konduktiva_view.rs‘ (Created — 556 lines): Implemented ‘KonduktivaNoteTrigger‘, ‘QuantizedRhythmRung‘, ‘KonduktivaLaneRhythmRow‘, ‘KonduktivaTelemetryState‘, ‘build_rhythm_map_ladder‘, ‘compute_variation_depth‘, ‘render_konduktiva_rhythm_visualizer‘, and 3 unit and headless ‘egui‘ tests.

- ‘src/lib.rs‘ (Modified): Exported ‘pub mod konduktiva_view;‘.
- ‘js/index.js‘ (Modified): Re-exported ‘./konduktiva-bridge.js‘.
- ‘js/nkeditor-sdk.test.js‘ (Modified): Added automated UDP loopback OSC test verifying ‘KonduktivaBridge‘ ladder rung selection, mute mask propagation, beat-window note trigger firing, and ‘/nkeditor/konduktiva/lane//rhythm‘ + ‘/trigger‘ packet delivery.
- ‘reports/report-015.md‘, ‘reports/report-015.html‘, ‘reports/report-015.pdf‘ (Created): Epoch 015 report in Markdown, grayscale HTML, and PDF formats.

Test coverage (69 total)

QA verdict

PASS

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